

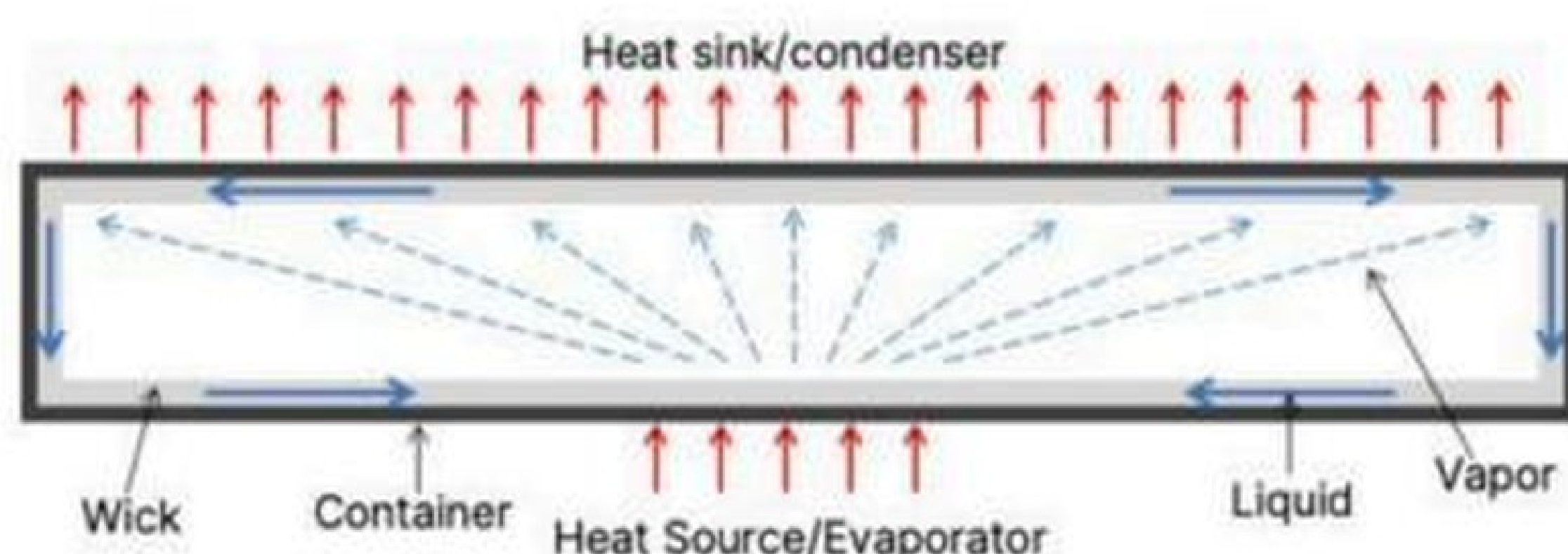
Capillary Action in Additively Manufactured Branching Channels

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Motivation

- Heat pipes and vapor chambers cool advanced electronics by moving heat through passive capillary wicks.
- Additive manufacturing can print channel geometries conventional fabrication cannot, supporting domestically manufactured thermal solutions.

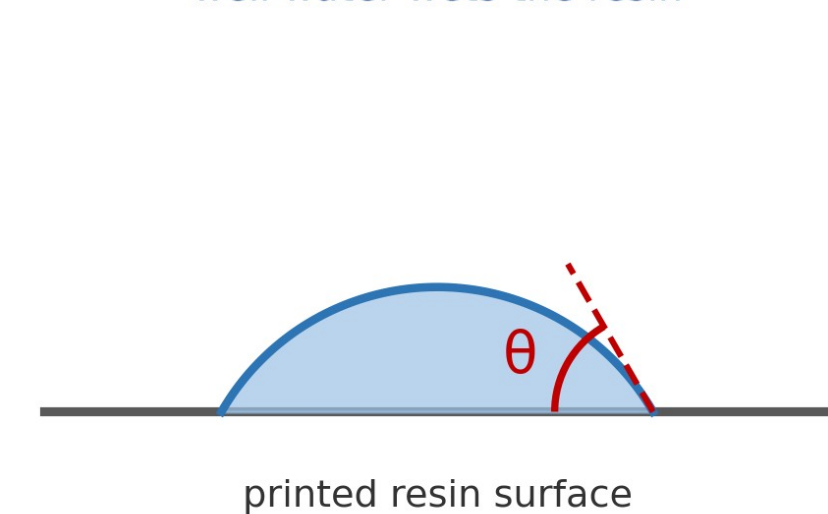


Vapor chamber operating principle: the wick returns liquid by capillary action. Image: Velardo et al. (2025), Thermal Science and Engineering Progress.

Objective: does branching network topology improve passive capillary transport? We compare Murray's Law, da Vinci's rule, and Hack's Law head-to-head in MSLA-printed capillary networks.

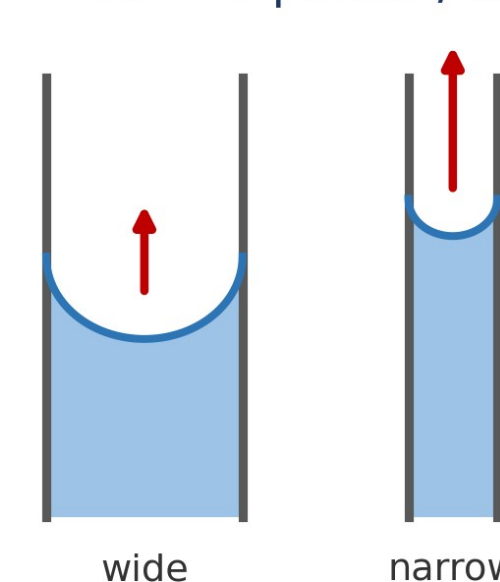
Governing Physics

Contact angle θ measures how well water wets the resin

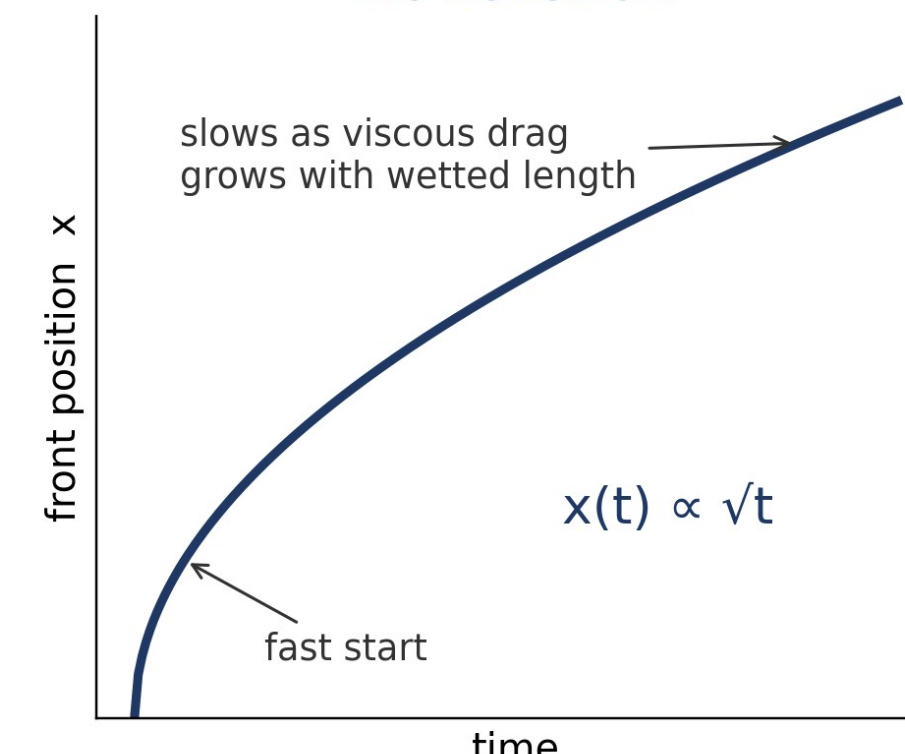


The curved meniscus pulls liquid in — narrower channel, stronger pull

$$\Delta P = 2\gamma \cos\theta / d$$



The front decelerates as it fills the network



γ = surface tension · θ = water-resin contact angle · d = channel diameter

Branching Design Laws

Murray's Law $d_0^3 = d_1^3 + d_2^3$

branch diameters that minimize viscous flow resistance

da Vinci's rule $d_0^2 = d_1^2 + d_2^2$

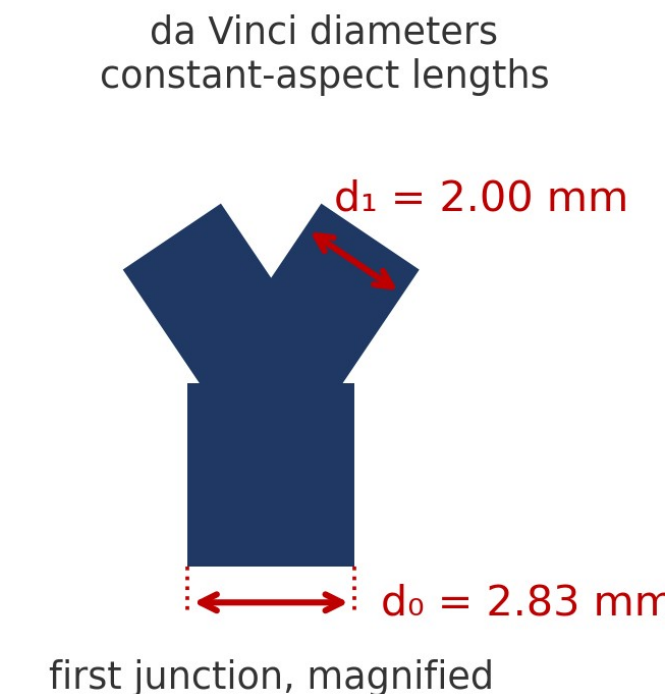
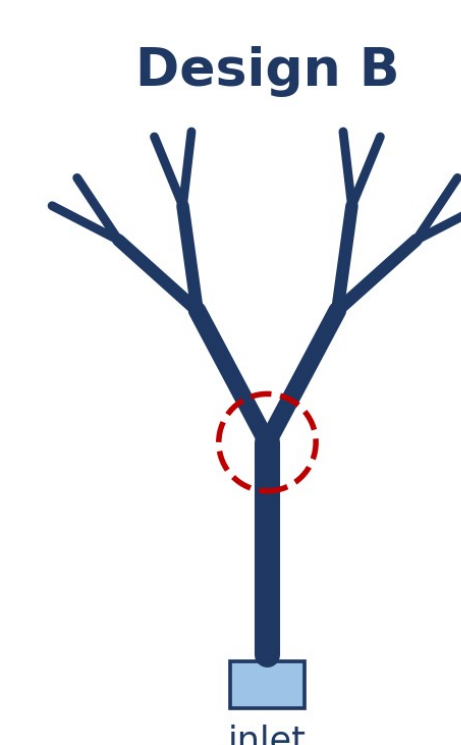
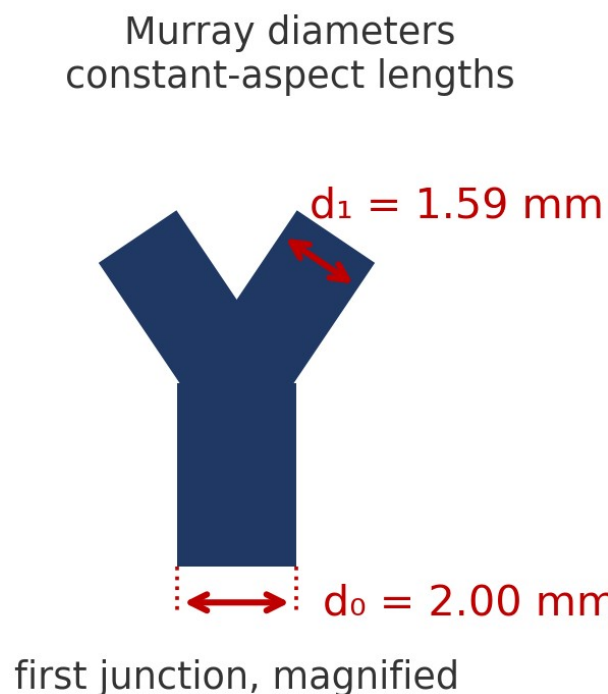
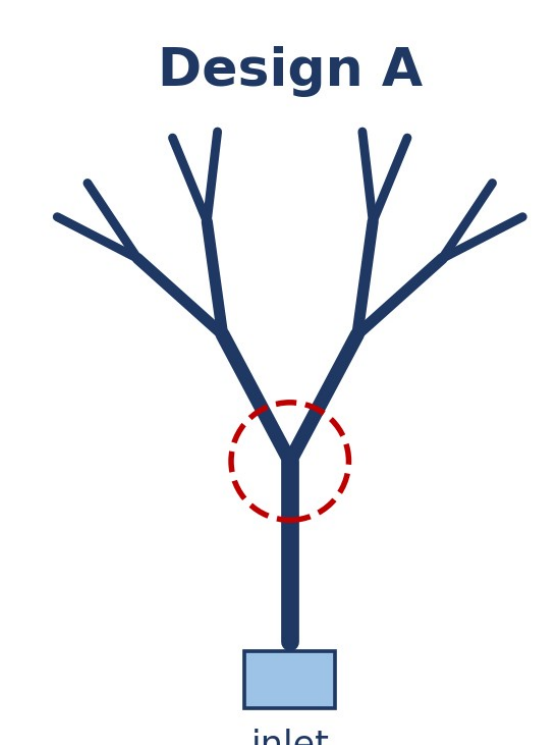
branch diameters that conserve total cross-sectional area

Hack's Law $L \propto A^h$, $h \approx 0.586$

channel length grows with the area it feeds

Murray / da Vinci set diameters; Hack sets lengths — independent choices, testable separately.

Channel diameters drawn to scale — tips fixed at 1 mm

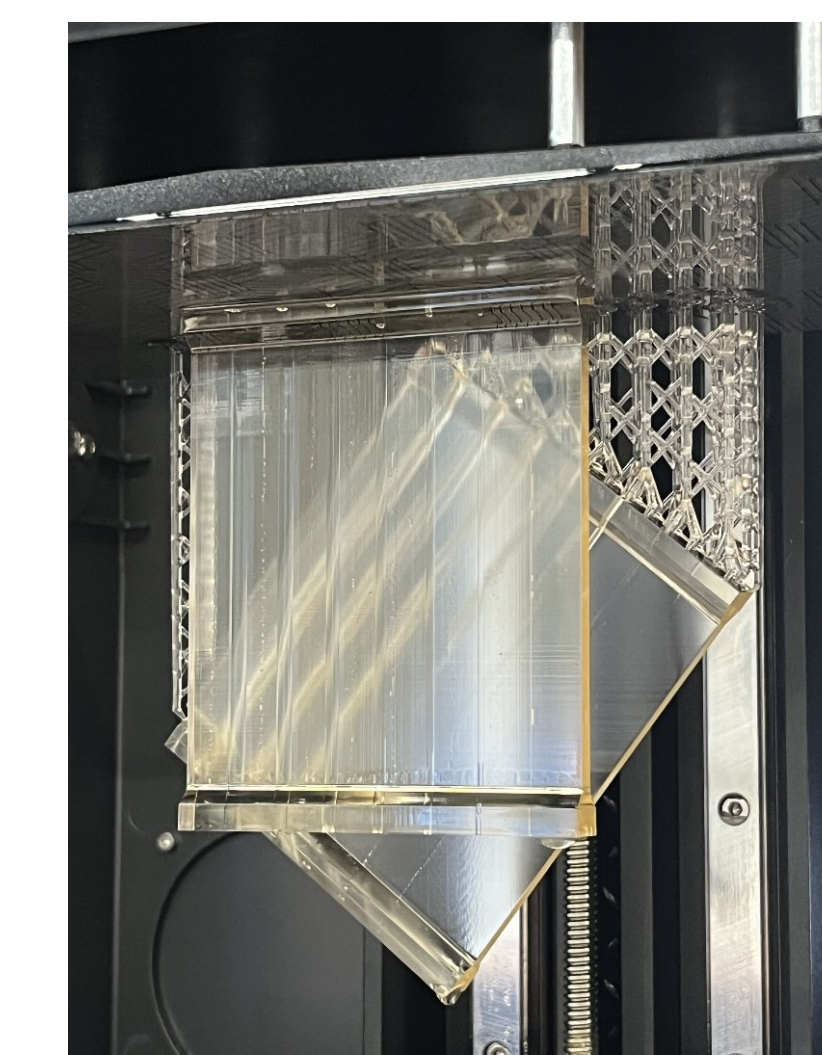


Comparing the Designs

- A vs B — same lengths, different area conservation laws.
- Hack's Law length scaling will be added in a future design to isolate the length law.

Fabrication

- MSLA: Elegoo Saturn 4 Ultra 16K with transparent 405 nm photopolymer resin.
- Enclosed channels printed directly into the plate — no assembly or bonding.
- 1 mm minimum channel diameter: smaller channels trap partially cured resin during printing, so tips are set at 1 mm and the network scales up from there.



The Experiment

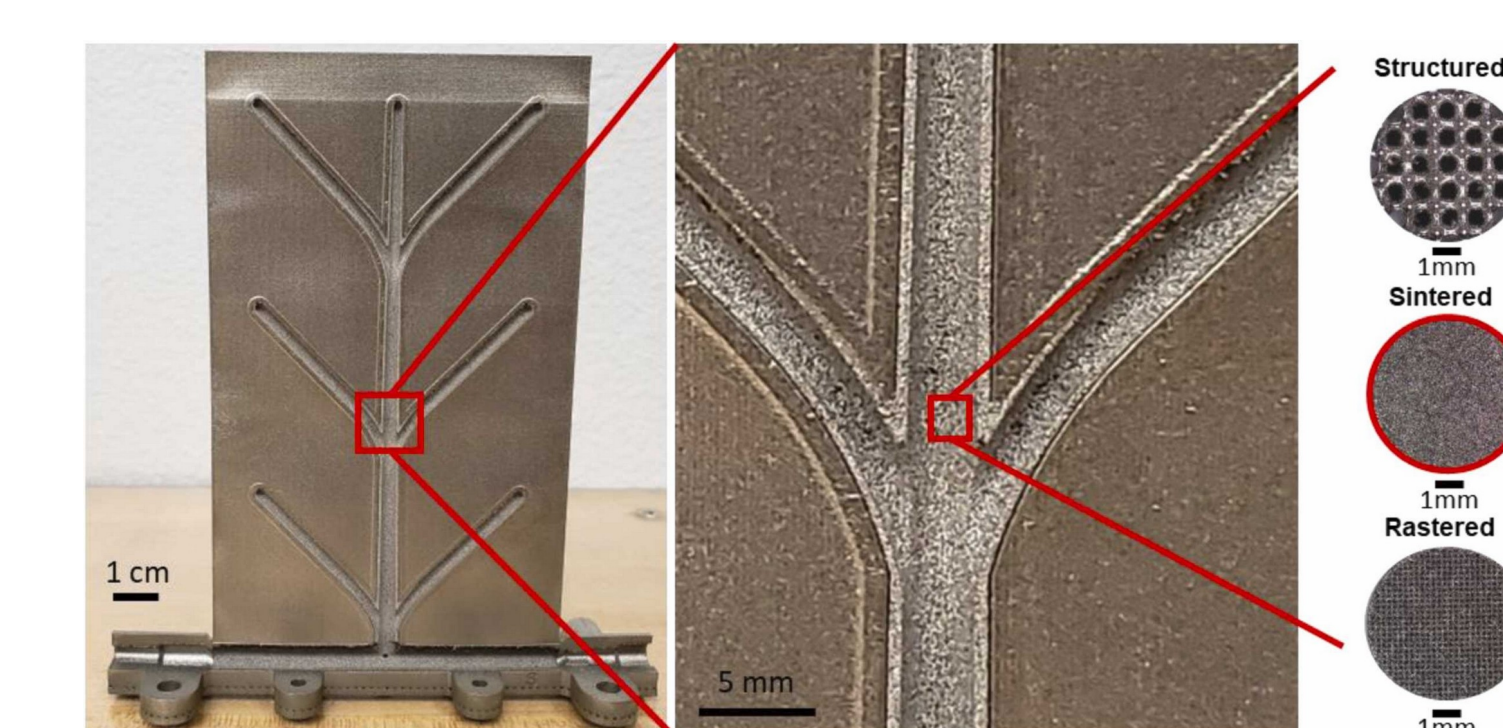
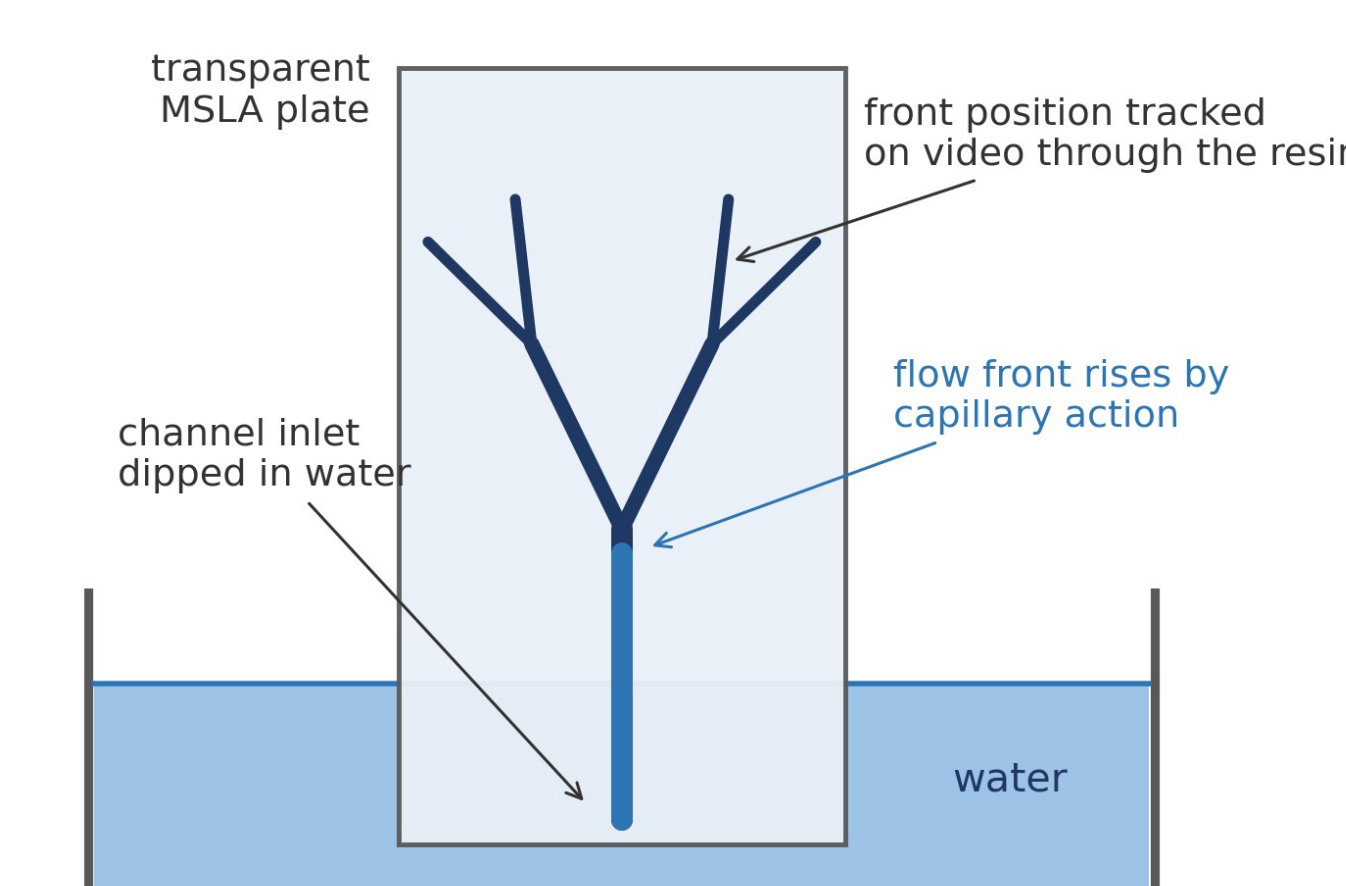


Image: Noe et al. (2025), Additive Manufacturing 99, 104669.

Path Forward

- Print and flow-test Designs A-C; compare how much, how fast, and how far water is transported.
- Measure the water-resin contact angle with matching print settings and post-cure.
- Examine the internal structure of printed channels to verify as-built geometry and surface quality.
- Extend the best-performing topology toward heat-pipe and vapor-chamber wick geometries.

Acknowledgements

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